



**Cambridge Instruments
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

RELAY TEST SET · THREE PHASE

RTS-33

Three-phase, two-unit relay test set — 0–360° phase angle, 0–50 A current, 415V 3-phase input. For distance, directional, and differential relay testing.

0–360° PHASE

Three-Phase Relay Test Set

A three-phase relay test set for testing distance, over-current, over/under-voltage, earth fault, and directional-element relays — a two-unit system (current/voltage source + a dedicated phase-shifter unit) giving full 0–360° phase angle control between voltage and current, for genuine directional and distance relay testing not possible with single-phase sets.

- **0–360° phase angle** — coarse 60° steps + fine $\pm 30^\circ$
- **0–50A across 5 taps** — 0–100% regulation each
- **Phase-to-phase fault mode** — selected phase currents in opposition
- **Digital timer** — 0–9.999s or 0–99.99s, two ranges
- **Phase sequence check** — built-in, before testing

Manufactured in India since 1963 — trusted by MSEB, Indian Railways, WBSETCL, etc.



- P** **0–360°** Phase angle, coarse + fine control
- A** **0–50A** 5 continuously variable taps
- V** **415V, 3-Phase** Input, 250 VA burden

01 Full Specification Sheet

PARAMETER	SPECIFICATION
Input	415 V $\pm 10\%$, 3 phase, 4 wire, 50 Hz A.C.
Burden	250 VA (current and voltage circuits)
Test Current Taps	0–1/5/10/25/50A (0–100% regulation per tap)
Duty	Continuous up to 10A; 25A & 50A: intermittent (5 min ON, 10 min OFF)
Test Voltage	0–110% of 110V ph-ph (63.5V ph-N), 3-phase, continuously variable
Fixed DC Voltage	Isolated 30V / 110V / 220V at 100 VA, available for 30 min
Phase Angle	0–360° – coarse 60° steps + fine $\pm 30^\circ$ (Unit 2, static phase-shifter)
Phase Angle Meter	360° range, 1° resolution
Time Measurement	0–9.999 sec (0.001s res.) and 0–99.99 sec (0.01s res.)
Current Metering	CT-operated ammeter per phase, 0–1A, 0–100% scaled, $\pm 1\%$ accuracy
Voltage Metering	0–200V, $\pm 1\%$ accuracy, 0.1V resolution, phase selector
Construction	Two-unit system; 1m interconnection cable; 4-core 5m mains cable
Warranty	12 months against manufacturing defects

02 Typical Applications

- Distance relay commissioning
- Directional and differential protection relay testing
- Over-current, earth fault, over/under-voltage relay testing
- Phase-to-phase fault simulation
- Indian Railways traction protection relay testing

03 What's in the Box

- ✓ Operation manual
- ✓ Schematic circuit diagram
- ✓ Works test report
- ✓ 1m interconnection cable (Unit 1 to Unit 2)
- ✓ 4-core, 5m mains input cable

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reliability since 1963.